Title: Comprehensive cost-effectiveness analysis of resective epilepsy surgery for adults in a National Health Service setting

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Abstract

Background

Epilepsy is a common, serious neurological condition affecting over 50 million people worldwide. Approximately 70% of those with epilepsy will be rendered seizure-free with a first or second appropriate anti-seizure medication. The remaining 30% of individuals are refractory to medications despite ongoing advances in the pharmacological armamentarium. In people with drug resistant focal epilepsy, particularly those with substrates for seizures identifiable on brain imaging, the most effective treatment is resective surgery.

Whilst the importance of progressing those with drug resistant focal epilepsy through an epilepsy surgery programme is clinically accepted, the cost-effectiveness of this approach is uncertain. Therefore, in developing the current National Institute of Health and Care Excellence (NICE) Guidelines on Epilepsies in children, young people and adults (NG217), we conducted a cost utility analysis of resective epilepsy surgery in adults.

Methods

Data were obtained from UK epilepsy surgery centres to evaluate the costs of pre-operative assessment. A de-novo decision analytic model, comprising of a one-year decision tree and life-time Markov model, compared resective epilepsy surgery to medical management from a national healthcare and personal social services perspective. According to NICE pre-requisites, an incremental cost-effectiveness ratio (ICER) below £20,000 cost per Quality Adjusted Life Year (QALY) was considered cost effective.

Findings

Data from 762 patients informed pre-operative evaluation costs, which were £8,182 per person on average. The total lifetime cost of epilepsy treatment for people that had surgical

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treatment was £56,911, compared with £32,490 for medical management. Total QALYs per person for surgery were 15·91 and 13.76 for medical management. Resective epilepsy surgery was shown to be cost effective with an ICER of £11,348 per QALY gained.

Interpretations

The study informs and strengthens recommendations to prioritise referral of those with drug refractory epilepsy to surgical centres, providing economic rationale for development and support of resective epilepsy surgery programs across national healthcare systems.

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Introduction

Epilepsy is a complex disorder that has profound impacts on an individual's quality of life, employment prospects and risk of premature mortality.[1,2] A symptom of many underlying causes, epilepsy is a common condition, with lifetime prevalence estimated at 7·6 per 1000 persons,[3] and contributes to a significant global burden of disease.[4,5] Drug refractory epilepsy is when two or more appropriate anti-seizure medications (ASMs) do not control seizures,[6] and this comprises around 20-40% of patients with epilepsy.[7] Drug refractory epilepsy results in increased risks of morbidity and premature mortality[8] and accounts for approximately 80% of healthcare costs in epilepsy care.[9] Direct costs include ongoing trials of ASMs, accounting for an estimated 50% of the total cost,[10,11] and, for example, hospitalisations owing to recurrent seizures or injuries. Indirect costs of epilepsy, for example unemployment or productivity costs – resulting from cognitive and psychosocial impacts– are several-fold greater than the direct care costs.[12] Also, as epilepsy associates with an increased risk of premature mortality, marked additional productivity costs accrue owing to potential lost earnings.

In people with drug-resistant epilepsy, arising from one specific part of the brain (focal epilepsy) resection of the epileptogenic zone can be considered. Epilepsy surgery has established clinical effectiveness for adults and children,[13–15] with up to 58% of treated individuals achieving seizure freedom.[15] A lower burden of seizures, particularly generalised tonic-clonic (convulsive) seizures, reduces the risk of sudden unexpected death in epilepsy (SUDEP).[16] Added benefits from improvements in seizure control include better health related quality of life in multiple domains including reduced depression and anxiety.[17]

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Despite suggested early referral for drug resistant epilepsy surgery,[18,19] there is marked under-utilisation across healthcare settings.[20–22] The average time to resective temporal lobe surgery from onset of seizures can be over 20 years.[23] It is estimated that over 10 million people globally are potential surgery candidates,[24] yet even in high-income countries only an estimated 1% of appropriate people are evaluated.[25] In more vulnerable groups, for example people with learning disabilities,[26,27] and particularly in resource underprivileged environments where the risks from seizures are greater, there is an even larger 'epilepsy surgery gap' (panel 1).

Multiple factors may contribute to the low rate of referral for epilepsy surgical evaluation.[18] These include perceived low rates of referred individuals proceeding to resection and evaluation costs.[28] Pre-surgical assessment does have significant upfront costs as it typically includes one or more video electroencephalogram (video EEG) admission, neuroimaging, neuropsychological evaluation, neuropsychiatric assessment and multidisciplinary input.[13,29] As such, constrained healthcare budgets are a prominent factor in limiting the referral and evaluation for surgery.[30]

While previous analyses have suggested that epilepsy surgery is cost effective, such studies have tended to rely on single centre data, [28,31–34] or only evaluated people who proceeded to resection, introducing probable positive bias.[35] To comprehensively address the cost effectiveness of epilepsy surgery over a life-time horizon, we analysed data across a state-funded (English) national health care system, estimating lifetime quality-adjusted life years (QALYs) and health service costs for all adults referred to resective epilepsy surgical programmes.

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Methods

Our model was developed as part of the NICE Guideline on Epilepsies in children, young people and adults NG217.[36] Model structure, inputs and results were presented to, and discussed with, the NG217 guideline committee for clinical validation and interpretation.[36] In instances where no data were identified, committee estimates were used to populate the model. The committee consisted of UK-based experts in epilepsy, epilepsy surgeons, neuropsychologists, pharmacists, and people with lived experience of epilepsy (appendix 1).

Model structure

We developed a decision analytic model comparing resective epilepsy surgery to medical management (MM) in adults from a United Kingdom National Health Service (NHS) and personal social services (PSS) perspective.

A cohort of adults with epilepsy was initially followed in the model through a one-year decision tree (figure 1). For the surgical arm, the decision tree modelled the costs and effects of pre-operative assessment and resective epilepsy surgery. The cost of pre-operative evaluation was included for those people who underwent resective epilepsy surgery and for those who underwent pre-operative evaluation but did not progress to resective epilepsy surgery (i.e., continued to receive MM). The MM arm modelled the cost and effects for one year of MM for people with drug refractory focal epilepsy. At the end of one-year, proportions of the cohort were seizure free, not seizure free, or dead.

Long-term outcomes were modelled by 49 one-year Markov cycles (thereby covering ages 36 to 85 years; figure 2). The states of the Markov model were 'seizure free for one year',

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‘seizure free for two years’, ‘seizure free for three or more years’, ‘seizure free off ASMs’, ‘not seizure free’, and ‘dead’. Within a one-year cycle, people were attributed an annual probability of remaining in their current health state, dying, or transitioning to ‘seizure free’ or ‘not seizure free’ – dependent on their original health state. Time-dependency was incorporated to track how long people were seizure free, and to capture the probability of discontinuing ASMs once seizure freedom was obtained.

Model inputs

The probability of obtaining seizure freedom in year one was derived from two randomised controlled trials (RCTs),[14,15] that were identified in the clinical review conducted as part of NICE’s guideline update.[36] The starting age of the model cohort was 35 years – in line with the average age of people included in the larger RCT.[15] The proportion of males in the model was 46.7%. This proportion was obtained from the long-term outcome study for resective epilepsy surgery,[37] as this study reported on a significantly larger cohort size in comparison to the rest of our outcome data and was based on UK outcomes. The probability of surgical mortality was also obtained from this dataset.[37] Probability of a permanent complication resulting from resective epilepsy surgery was 4.0%.[38–41] Data inputs for the one-year decision tree are presented in table 1. Owing to an absence of long-term RCT data relating to epilepsy surgery, outcomes for surgery and MM were obtained from two long-term observational studies (table 1).[37,42] Transition probabilities were calculated for the risk of relapse (transitioning from ‘seizure free’ to ‘not seizure free’) and remission (transitioning from ‘not seizure free’ to ‘seizure free’). Yearly probabilities for relapse and remission in the MM arm were calculated from the cumulative probabilities previously reported,[42] assuming a constant rate of remission and relapse over the five-year period. Transition probabilities in the surgical

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arm were calculated differently to those in the MM arm because annual data were available for the number of people entering remission and relapsing (up to year 15). These data were used to calculate individual yearly probabilities up to year 15 and were subsequently extrapolated beyond this for the remainder of a person's lifetime (appendix 2). The committee assumed a lifetime 4% probability of re-operation for those who had previously undergone resective epilepsy surgery.

Standardised Mortality Ratios (SMRs) for 'seizure free' and 'not seizure free' individuals were applied to the general population mortality rates.[43] The SMR for 'not seizure free' and 'seizure free' MM were obtained from published estimates.[38–41,44] The SMR for 'seizure free' following epilepsy surgery was adjusted to account for the differing definitions of seizure freedom in our two long-term outcome studies (appendix 3).[37,42]

Utility values used are presented in table 2. Health state utilities were based on reported values for people who were 'seizure free'; 'people experiencing a ≥50% reduction in seizures'; and 'people experiencing a <50% reduction in seizures.[45] The utility for 'seizure free' in the surgical arm was adjusted to account for the different definitions of seizure freedom (appendix 4). To obtain a utility value for 'not seizure free', data on the proportion of people experiencing a ≥50% reduction in seizures, and people experiencing a <50% reduction in seizures,[46] were multiplied by published utility values.[45] A yearly utility decrement of 0.2 was applied for those who experienced long term complications from surgery.

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Pre-operative assessment resource use for individuals undergoing pre-surgical evaluation were collected from adult surgical centres in England and Wales using a standardised data capture form (appendix 5). Each centre recorded data for a minimum of 50 consecutive patients who had completed surgical work-up. Data were anonymised and analysed in aggregate to obtain the mean number of pre-surgical evaluation tests per patient. The cost of pre-surgical evaluation was calculated by multiplying the mean number of tests by the unit cost and summing these (table S6; appendix 6).

Unit costs used in the model were obtained from published sources specific to the NHS.[47,48] Published unit costs were unavailable for amytal testing, magnetoencephalography and electrocorticography, and therefore obtained from the participating surgical centres that provided these tests. A number of costs used in the model were updated for this publication and so differ slightly from those published in the NICE guideline (appendix 7).[36]

Potential adverse consequences of epilepsy surgery include immediate complications such as infection or haemorrhage, which can usually be treated promptly and have no long term sequalae. Also, there are small risks of stroke and mortality. The average long-term cost of surgical complications, this encompassing all potential surgical complications, was taken to be £5,000 per year over a lifetime. This figure was thought likely an over-estimate by the NICE Committee but retained to avoid positive bias in the model.

The cost of outpatient contacts included the cost of an initial neurology appointment, subsequent neurology appointments and primary care consultations (table S8; appendix 6). Inpatient and emergency admission costs were also included in the analysis (table S9;

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appendix 6). Probability of service use for the costs of outpatient contacts and admissions was obtained from previous literature,[49] and informed by expert committee opinion. The NICE committee estimated the proportion of people who would receive each ASM and assumed people with drug resistant epilepsy would receive an average of 2.5 ASMs. The total annual cost of ASMs per person was calculated to be £1,082 (table S10; appendix 6). The cost of the resective operation was £10,185.[47]

Analysis

To calculate overall cost effectiveness, costs and QALYs for each cycle were calculated by multiplying the proportion of the cohort in each state by the corresponding cost or utility, with a half-cycle correction applied. Costs and QALYs were discounted to reflect time preference (3.5%). Costs and QALYs were each summed across the lifetime horizon (50 years). The incremental cost-effectiveness ratio (ICER) was then calculated by dividing the difference in summed costs associated with surgery and MM by the difference in summed QALYs.

The model was run probabilistically using Monte Carlo simulation to take account of the uncertainty around input parameter point estimates. A probability distribution was defined for most model input parameters. When the model was run, a value for each input was randomly selected simultaneously from its respective probability distribution; mean costs and mean QALYs were calculated using these values. The model was run 10,000 times. Main results of the model are presented probabilistically. One-way sensitivity analyses are presented deterministically.

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For the probabilistic analysis a beta distribution was applied to the following data inputs: probability of not being seizure free for MM, probability of mortality in the surgery arm, probability of relapse and remission for surgery and MM, and the probability of being a surgery candidate. A gamma or beta distribution was applied to the average number of pre-surgical evaluation tests. The distribution applied was dependent on the usage of each specific test. When the average resource use per person was above one, a beta distribution was applied – a gamma distribution was applied to remaining tests (appendix 6). A gamma distribution was applied to the cost of surgery and the utility values used in our model (appendix 4). A log-normal distribution was applied to the risk ratio for seizures at one year (surgery versus waiting list) and the SMRs (table 1).

A total of 18 deterministic one-way sensitivity analyses were conducted (appendix 7). These included using utilities from different sources; altering the costs for surgery and pre-surgical evaluation; employing a 15-year time horizon; assuming people did not discontinue their ASMs once they obtained seizure freedom; and assuming a higher cost for certain pre-surgical investigations.

The model was systematically checked by the health economist undertaking the analysis (AB); this included inputting null and extreme values to check that results were plausible. The calculations were systematically checked by a second experienced health economist (DW).

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Results

Fourteen epilepsy surgical centres were contacted and ten provided data for a total of 762 adult individuals (appendix 8). The mean number of preoperative evaluation tests per person and corresponding unit costs for each test are presented in table S6; appendix 6. The total cost of preoperative assessment was estimated to be £8,182 per person.

Another key input – the proportion of people undergoing resective epilepsy surgery having completed preoperative evaluation – was also obtained from this survey. The probability of being a surgical candidate was 41·3% (315/762), which included those people who were eligible for resective epilepsy surgery and for whom surgery went ahead or was due to take place. This probability of 41.3% did not include those people who were eligible for surgery but did not consent to surgery.

For the probabilistic base case results, the total cost per person for surgery was estimated to be £56,911, and the total cost for MM was £32,490 (table 3). Total QALYs per person for surgery was 15·91 and MM 13·76. Overall, resective epilepsy surgery was found to be cost effective with an ICER of £11,348 per QALY gained (table 3). Deterministic results are listed in table S12; appendix 9.

The results of the probabilistic analysis are illustrated in figure 3, where each of the 10,000 iterations is plotted. Resective epilepsy surgery had a 97·0% probability of being cost effective at NICE's threshold of £20,000 per QALY gained (indicated by the proportion of iterations to the right of the dotted line). There was a 99·5% probability of surgery being cost effective at NICE's upper threshold of £30,000 per QALY gained (not shown in graph).

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In all deterministic sensitivity analyses, resective surgery was cost effective at NICE's £20,000 threshold per QALY apart from when the time horizon was reduced to 15 years (ICER: £28,093) and when the overall worst-case scenario was employed (table S13; appendix 9). The overall worst-case scenario comprised of all the scenarios tested in the one-way sensitivity analyses that favour MM to surgery (table S11; appendix 7). This sensitivity analysis was conducted to test these assumptions, but the NICE committee noted that the overall worst-case scenario was highly unlikely to be representative of clinical practice.

A one-way sensitivity analysis was undertaken assuming the highest cost of pre-surgical evaluation across all ten surgical centres (£13,178 compared to £8,182) – illustrating that surgery was cost-effective with an ICER of £16,679 per QALY gained. Another sensitivity analysis was conducted incorporating a higher cost for stereo-EEG (sEEG). This analysis assumed that 60% of people undergoing sEEG received standard sEEG (the NHS reference cost used in the base case analysis [£14,638]) and 40% of people received a more complex sEEG implantation (£39,577; costs obtained from two participating surgical centres). This resulted in a mean cost for sEEG of £24,613. The results of this analysis indicated that epilepsy surgery remained cost-effective at £12,889 per QALY gained. Results of all 18 deterministic one-way sensitivity analyses are provided in table S13; appendix 9.

Discussion

The clinical effectiveness of epilepsy surgery is established in appropriately selected cases. Despite this, resective surgery for people with drug resistant epilepsy is under-utilised.[13–15,20–22] Scarce budgets drive the need for cost-effectiveness analyses to support and expand epilepsy surgery programs.[50] To enable change, analysis of people referred for epilepsy

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surgery is essential – namely an evaluation of whether referral for epilepsy surgery is cost-effective irrespective of whether a given individual proceeds to resection.

Our nationwide multicentre pre-surgical evaluation survey included costs of all adults referred for pre-surgical evaluation, thereby reflecting real-world costs of an epilepsy surgery program. When taking this all-encompassing approach, we demonstrate epilepsy surgery for drug resistant epilepsy is cost effective 97·0% of the time, at NICE's threshold of £20,000 per QALY gained. As such, this study provides a more definitive economic rationale for health system funding for referral to epilepsy surgery programs. These data seem broadly applicable to other healthcare settings, although the economic rationale for resective epilepsy surgery in low income to middle income healthcare countries requires specific consideration (panel 1).

The process of pre-surgical evaluation itself is a significant proportion of the cost of a surgical epilepsy program. 58·7% of people in our cohort who underwent pre-surgical evaluation did not proceed to resection. Pre-surgical evaluation in an individual who does not proceed to surgery does, though, provide valuable insights to optimise patient care. Benefits of such assessment, which were not within the scope of our analysis, include uptake of alternative therapies, such as vagal nerve stimulation, and identification of psychogenic non-epileptic/functional seizures, which can facilitate appropriate reductions in ASM burden and onward referral to appropriate services.

The cost of pre-surgical evaluation may be greater in certain sub-groups. For example, people with intellectual disabilities can require additional healthcare professional involvement in the

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pre-surgical evaluation process, such as further investigations to identify the seizure onset zone, specialist work to prepare for sEEG and additional staff time for pre-surgical counselling and consent. Based on the one-way sensitivity analysis conducted assuming a higher pre-surgical evaluation cost – and assuming equal efficacy – it is highly likely surgery is cost-effective for this cohort of people. Further information is, though, required to determine the efficacy of surgery and the subsequent cost-effectiveness in this sub-group.

The cost of sEEG is variable depending on the complexities of a person's epilepsy. The cost of sEEG used in the base case analysis was obtained from NHS reference costs,[47] although the NICE committee acknowledged this cost may be an underestimate for more complex cases. Since this model was developed, the frequency with which sEEG is used in pre-surgical evaluation has increased in addition to an increase in costs. The sensitivity analysis conducted assuming a higher cost for sEEG likely covers the increase in costs but does not account for the increased frequency with which sEEG is deployed. Additional research is therefore required to determine the cost-effectiveness of resective epilepsy surgery with increased use of sEEG within the pre-surgical evaluation pathway.

Utility values in our model were obtained from a non-drug refractory population due to a lack of data in a drug-refractory cohort. Different studies have produced different utility values for 'seizure free' and 'non-seizure-free' health states in non-drug refractory epilepsy populations. Although, the utility values used in the model come from a relatively large UK study, several sensitivity analyses were conducted using different utility values to assess this uncertainty. The results of these analyses indicated the model was potentially sensitive to the utility values used, but, even under the most conservative assumption, the ICER was less than £20,000 per

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QALY gained (table S13; appendix 9). Also, epilepsy surgery can result in benefits even if people are not rendered seizure free, such as fewer seizures and seizures of lower severity. There can be consequent cost savings owing to fewer attendances at Emergency Departments or a reduction of ASMs. Were these factors to be included, the cost-effectiveness of resective epilepsy surgery would be increased.

The analysis did not include paediatric cases. The logistical organisation for epilepsy surgery is different for children in the United Kingdom where resective epilepsy surgery is carried out at four designated Children's Epilepsy Surgery Service (CESS) Centres. There may be certain additional costs in children, for example the potential need for imaging to be performed under general anaesthesia. It is, though, inferred that epilepsy surgery is likely to be more cost effective in children as earlier control of seizures offers the prospect of earlier drug reduction and decreased risk of mortality.[51] Although, health economic analyses at NICE take a health service perspective, rather than societal, it is important to note that earlier seizure control in children results in better socioeconomic and educational attainment.[52]

Limitations

There are several limitations to our study. Treatment effects in our analysis are based on two small RCTs, and therefore long-term outcomes were calculated using observational studies.[37,42] RCT evidence assessing the effectiveness of epilepsy surgery will likely always be short-term owing to ethical concerns associated with conducting longer-term RCTs – delaying epilepsy surgery when of proven benefit would not reflect clinical equipoise. Despite this limitation, one-year seizure freedom rates reported in the observational studies correlated well with the RCT data. We also conducted a sensitivity analysis using a 15-year

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time horizon that meant outcomes for surgery were not extrapolated beyond the observational study period for the surgery arm. In that sensitivity analysis epilepsy surgery was still found to be cost-effective at NICE's £30,000 threshold.

The average age for people entering our model was obtained from one[15] of the two RCTs[14,15]. The combined age from the two RCTs was not used owing to differentials in the average age and patient numbers observed in the smaller RCT [14] To overcome this limitation, we conducted a sensitivity analysis using only the larger RCT for our effectiveness data.[15] This showed that surgery was cost-effective with an ICER of £11,175 (compared to £10,932 in the deterministic base case).

Certain resource allocation assumptions were partly based on the expertise of the committee. For example, the cost of complications and the probability of reoperation. The cost of two preoperative assessment tests (magnetoencephalography and electrocorticography) were obtained from the surgical centres who provided these tests. For the committee estimates, there were a range of personnel (appendix 1) and the level of consensus was high. Owing to a lack of published data on the cost of fMRI, the committee assumed fMRI to be of the same cost as an MRI. Although the committee noted that this cost was likely an underestimation of the true cost, they agreed the cost would not alter the results of the cost-effectiveness analysis as use of this test was relatively low (table S6; appendix 6). In addition, the costs of preoperative assessment were tested in two sensitivity analyses (assuming a higher and lower cost) and the results were found to be robust (table S11; appendix 7 for details, and table S13; appendix 9 for results).

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It was difficult to be certain about the cost and quality of life impact of complications following resective epilepsy surgery, as this can vary markedly between individuals. By taking the cost of a permanent complication as £5,000 per year over a lifetime horizon, even if the rate of complications was higher than the 4% assumed in our model, the total cost of complications following epilepsy surgery was likely over-estimated.

Epilepsy surgery evaluation in adults and children, now involves increasingly complex procedures, including stereo-EEG recording. Single centre data show that costs for such procedures can be materially higher and, by definition, individuals proceeding to intracranial recording have more complex epilepsy and lower probabilities of post-operative seizure freedom.[53] Cost-effectiveness analysis of novel surgical techniques across health systems, deploying our methodology of evaluating all people referred for epilepsy surgery and not only those who proceed to resection, is recommended. Sub-group analyses could, for example, compare those who require intracranial recording to those who do not and evaluate cost-effectiveness by location of the resective surgery.

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Conclusion

Epilepsy surgery provides improved health outcomes for people with drug resistant epilepsy.[13–15] We demonstrate that being evaluated for epilepsy surgery in drug resistant epilepsy is cost effective from a national health service perspective even when considering all people with drug resistant epilepsy and not just those who proceed to resection. Confirming cost-effectiveness of referral for epilepsy surgery should offer increased support to development and delivery of epilepsy surgery programmes.

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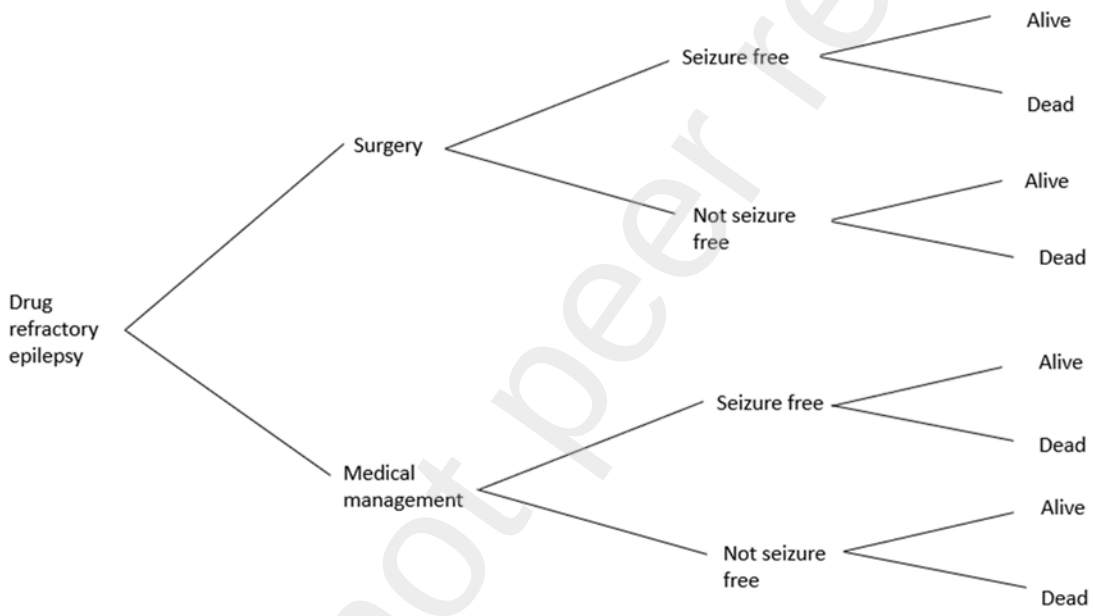
Comprehensive cost-effectiveness analysis of resective epilepsy surgery for adults in a National Health Service setting

Figure 1: One-year decision tree model



Binary outcomes in the epilepsy surgery pathway that reflect clinical decision making and possible outcomes are illustrated. In this one-year decision tree model evaluating resective epilepsy surgery and medical management, after one year people can either be alive or dead dependent on their seizure status.

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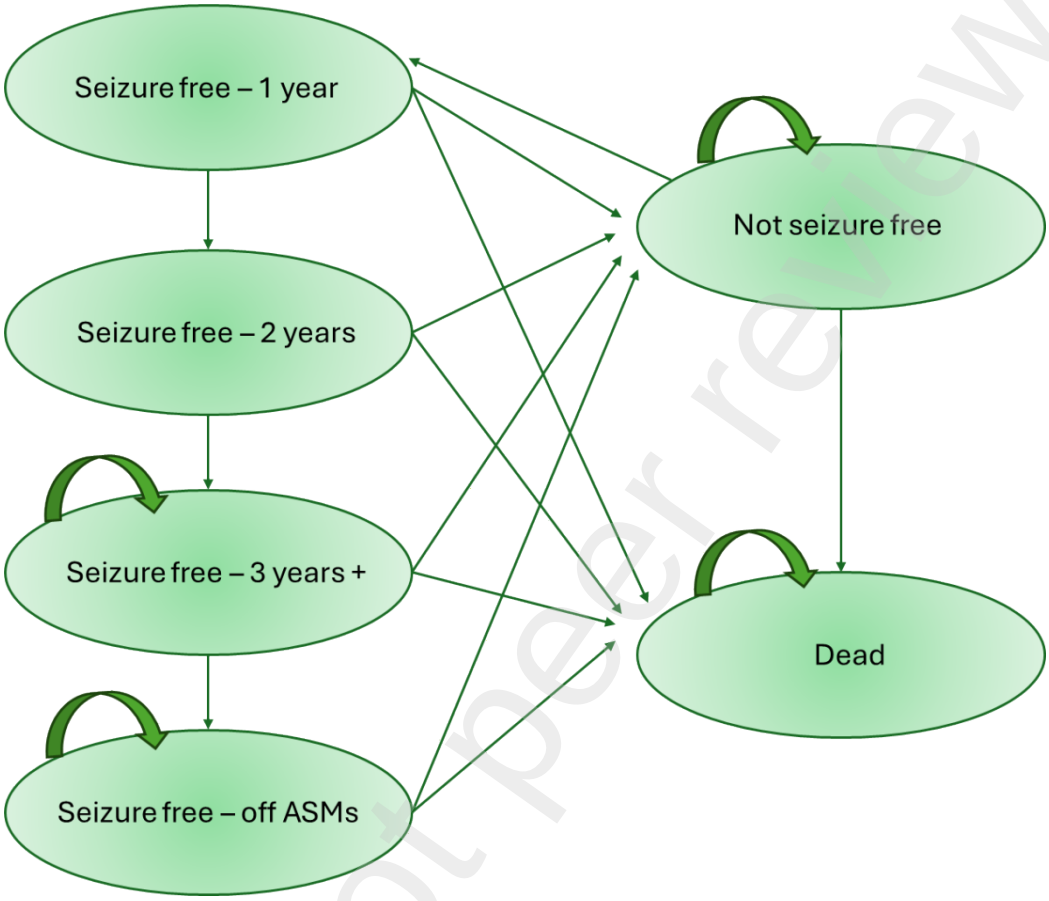
Figure 2: Long-term Markov model demonstrating possible health states in the epilepsy treatment pathway



A schematic of the long-term Markov model where people can be either, seizure for a given period of time (prescribed or not prescribed anti-seizure medications), not seizure free or dead.

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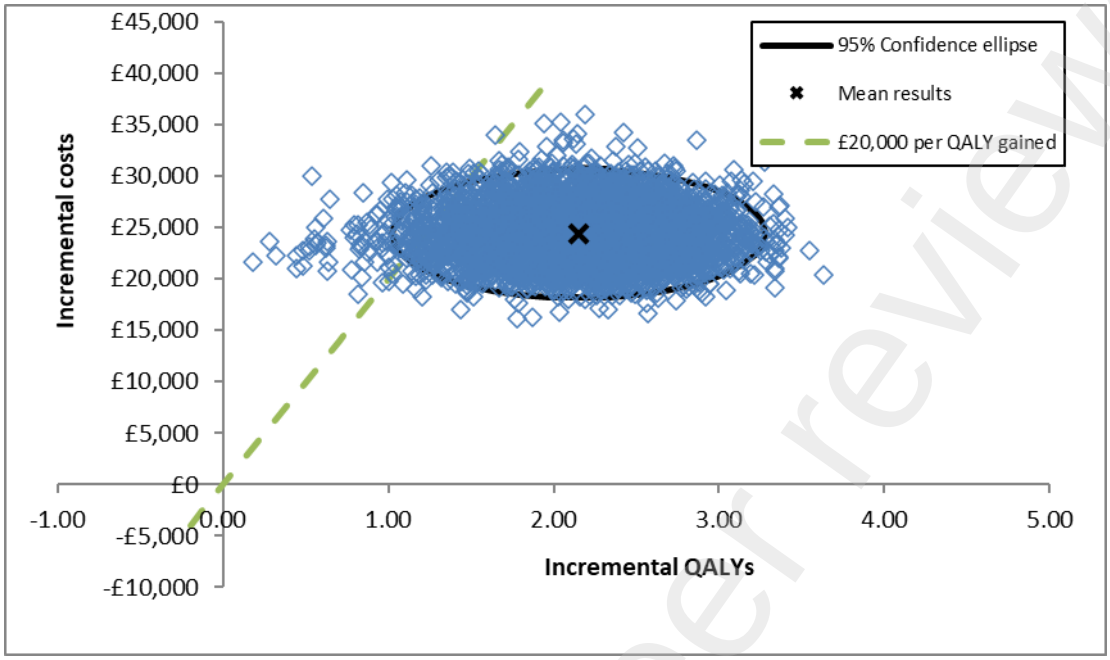
Figure 3: Cost-effectiveness plane



The cost effectiveness plane illustrates the results of the probabilistic sensitivity analysis, plotting each of the 10,000 iterations. Overall resective surgery had a 97% probability of being cost effective at the standard NICE threshold of £20,000 per Quality Adjusted Life Year gained.

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Tables

Table 1: Model inputs: clinical outcomes. MM=medical management. ASM=anti-seizure medication. SMR=standardised mortality ratios. LnRR=log response ratio, SE=standard error, NA=not applicable.

Data input	Base case value	Probability distribution
One year decision tree outcomes		
Probability of 'not being seizure free' in the MM arm	96·7% [14,15]	Beta (Alpha: 59, Beta: 2)
Risk ratio	0·42 [14,15]	Lognormal (LnRR: -0.87, SE: 0·16)
Probability of mortality in the surgery arm	0·77% [37]	Beta (Alpha: 5, Beta: 644)
Probability of long-term complication from resective epilepsy surgery	4·0% [38–41]	NA
Long-term Markov model outcomes		
Annual probability of remission surgery	Various annual probabilities (see appendix 2) [37]	Beta
Annual probability of relapse surgery	Various annual probabilities (see appendix 2) [37]	Beta
Annual probability of remission MM	5·6% [42]	Beta (Alpha: 62, Beta: 184)
Annual probability of relapse MM	22·0% [42]	Beta (Alpha: 42, Beta: 17)
Annual probability of discontinuing ASMs each year $\geq$ 3 years of seizure freedom	15·7% [54]	NA
Probability of reoperation	4·0% (Committee assumption)	NA

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SMR 'seizure free' surgery	2·42 (see appendix 3)	Made probabilistic based on the probability distributions applied for the SMR of seizure free MM and the SMR of not seizure free.
SMR 'seizure free' MM	1·78 [44]	Lognormal LnRR = 0·575, SE = 0·678
SMR 'not seizure free'	5·40 [38–41]	Lognormal LnRR = 1·686, SE = 0·158

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Table 2: Model inputs: utility values.

Data input	Base case value	Probability distribution
'Seizure free' surgery	0·858 [37,45]	See appendix 4
'Seizure free' medical management	0·869 [45]	See appendix 4
'Not seizure free'	0·689 [45,46]	See appendix 4
Annual utility decrement for long-term complications of surgery	-0·2 [38–41]	NA

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Table 3: Cost effectiveness results per person (probabilistic results). QALY=Quality-adjusted life-years.

.	**Surgery**	**Medical management**	**Surgery minus medical management**
Assessment for resective surgery cost	£20,823	£0	£20,823
Resective surgery cost	£10,201	£0	£10,201
Outpatient appointment cost	£3,631	£5,517	-£1,887
Anti-seizure medication cost	£14,522	£20,022	-£5,500
Admission cost	£3,254	£6,951	-£3,697
Reoperation costs	£678	£0	£678
Complications cost	£3,804	£0	£3,804
All costs	£56,911	£32,490	£24,442
QALYs	15·91	13·76	2·15
Incremental cost per QALY gained			£11,348

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Panels

Panel 1: Applicability to other health care settings.

Epilepsy disproportionately affects people from lower socio-economic groups. Despite this, even in high-income countries, the number of people from poorer backgrounds referred for specialist care is lower than from richer deciles resulting in both diagnostic and treatment gaps.[55] In low to middle income countries (LMICs) such gaps are much larger, the risks from seizures greater (for example owing to cooking on open fires; washing clothes in rivers – situations in which seizures are much more likely to result in harm) and stigmatisation of epilepsy can be very marked.[4] Inconsistency in the supply of medication also associates with specific hazard. 'Stock outs' may mean that people are suddenly without ASMs for several months resulting in worse and more frequent seizures for those with drug refractory epilepsy as well as associated morbidity and mortality.

Epilepsy surgery, it could be argued, is therefore more needed in LMICs as it offers the possibility of seizure remission and, potentially, less need for ASMs.[56] Although there are limited data on epilepsy surgery in LMICs, favourable outcomes are reported.[57–60] Certain centres in LMICs are also able to offer work up of complex cases, including intracranial recording, again demonstrating positive outcomes,[57,61] whilst others have developed models of data sharing to enable consensus treatment decisions.[58,62]

Relatively few data are available on the costs of resective epilepsy surgery in LMICs. In Panama, for example, the cost was estimated at USD 9,850 per patient although this included intraoperative corticography as well as invasive intracranial recording.[62] An Indian team identified appropriate surgical candidates using the minimal required tests of video-EEG and 1·5 Tesla MRI scans and estimated a total cost (investigations and the operation itself) of Rs

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92707 (USD 1,324).[63] In this cohort Engel Class 1A outcomes (completely free of seizures following epilepsy surgery) were observed in 92·5% of people with more than one year of follow up.[63]

Overall, costs for epilepsy surgery are substantially less than in high-income settings with potentially greater benefits.[64] Although costs are lower in LMICs, these countries will typically have a smaller budget for health care and therefore cost-effectiveness thresholds will be lower. Our model illustrated that resective epilepsy surgery is highly cost-effective providing a clear mandate that appropriate individuals are considered for resective epilepsy surgery in LMICs.

Panel 2: Research in context.

Evidence before this study

We searched PubMed for English language publications from January 1st 1980 to Jan 1st 2024, with the terms: [“cost-effectiveness” OR “cost analysis” OR “economic evaluation”] AND [“epilepsy surgery” OR “temporal lobectomy”]. Whilst cost-effectiveness analyses have previously supported the economic rationale for resective epilepsy surgery for drug refractory epilepsy, most studies have focused on single centre data in a North American context. Several studies from the 1990s have not integrated data from landmark randomised control trials that provide clinical effectiveness data, whilst others do not evaluate the lifetime costs and effects of epilepsy surgery. No recent studies have assessed the cost-effectiveness of resective epilepsy surgery at the point of referral to surgical centres, thus omitting the costs and the outcome of pre-surgical evaluation.

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Considering a nationwide healthcare system perspective, a single study has been published in a European context supporting the economic rationale of resective epilepsy surgery. That work, though, evaluates the economic rationale without including costs of presurgical evaluation if a patient is deemed inoperable (for more information on these studies see supplementary material table S14). As such our study offers unique perspectives through the utilisation of national data on the cost-effectiveness of resective epilepsy surgery, including the significant costs of all people undergoing pre-surgical clinical evaluation – also including the costs of those who are evaluated but who do not proceed to resection.

Added value of this study

Here, total costs of resective epilepsy surgery programs in adults were evaluated with a lifetime Markov model. The study demonstrates cost-effectiveness of resective epilepsy surgery programs. Distinct from previous work, our study provides evidence of cost-effectiveness from a total healthcare perspective rather than only considering cost-effectiveness in people who have already been fully assessed as operable.

Implications of all the available evidence

Resective epilepsy surgery for drug resistant epilepsy is cost-effective, even when including assessment of people who are subsequently found not to be appropriate for operative intervention. The study provides increased support for referral to epilepsy surgery programs.

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Contributions

The manuscript was conceptualised by AB, PK and AS with additional input throughout from SKB and DW. PK, JdT, OS, ISZ, JW, KS, VC, BZ, HF, SS, KH, LK, NM, EMA, SE, RC were involved in data collection from surgical centres. AB and DW oversaw all health economics analyses. AB, SKB, DW, PK and AS drafted and revised the manuscript with comments from all co-authors incorporated.

Declarations of interest

The authors declare no direct conflicts of interest

Data sharing

Raw data are available on reasonable request.

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